

5-10-15-20...40. Q9 output is used for displaying 45th minute. The cathode of the 45th-minute LED is connected to the output of OR gate N3 of IC CD4071 (IC10), which is normally low until 50th-minute input pulse is received. Q9 output is also connected to one of

the two inputs of AND gate N7 of IC CD4081 (IC9). The other input of N7 is connected to positive terminal of the supply. So whenever Q9 goes high, the output of N7 also goes high. This makes pin 13 of IC7 high to disable the counter. At the same time, the output

of N7 drives transistor T1 into saturation. Thus pin 13 of IC8 goes low to enable the counter and light up 50th- and 55th-minute LEDs at Q1 and Q2 outputs of IC8. The Q3 output of IC8 resets IC7 and IC8 at 60th pulse, causing Q0 output of IC7 to go high, and 60th-minute LED glows. In addition, it provides an output pulse every 60th minute (one pulse per hour (PPH)) for the next stage. For setting the minutes display, the counters can be reset to '00' by momentarily pressing push-to-on switch S2. SPDT switch S6 connects to switch S13 and press switch S13 until the desired minutes LED is 'on'.

Hours counter and display. This circuit is similar to the minutes counter

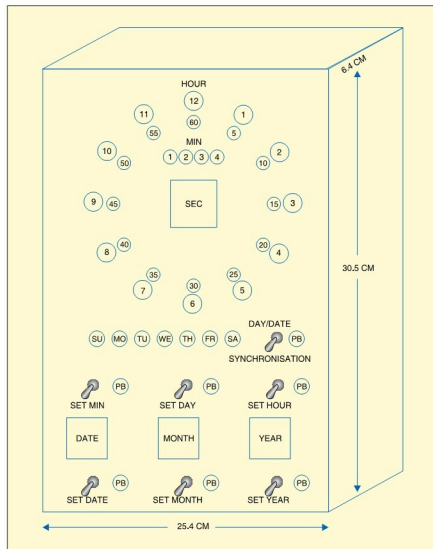


Fig. 2: Proposed analogue clock panel with mounting arrangement for LEDs and switches

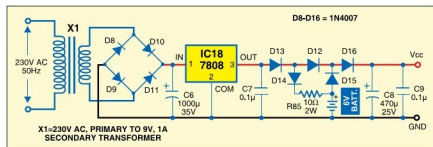


Fig. 3: Power supply circuit

PARTS LIST

Semiconductors:

- IC1 - CD4060 oscillator and counter
- IC2, IC6-IC8, IC11, IC12, IC16 - CD4017 decade counter
- IC3, IC4, IC14, IC15 - CD4033 counter and 7-segment decoder
- IC5 - CD4001 quad NOR gate
- IC9 - CD4081 quad AND gate
- IC10 - CD4071 quad OR gate
- IC13 - CD4027 dual J-K flip-flop
- IC17 - CD4069 hex inverter
- IC18 - 7808, 8V regulator
- T1, T2 - BC547 npn transistor
- D1-D7 - 1N4148 switching diode
- D8-D16 - 1N4007 rectifier diode
- LED1-LED35 - LT543 common-cathode 7-segment display
- LED1-LED35 - 5mm LED

Resistors (all 1/4-watt, ±5% carbon unless stated otherwise):

- R1 - 33-kilo-ohm
- R2 - 10-mega-ohm
- R3-R7, R9-R15 - 100-kilo-ohm
- R16-R79 - 680-ohm
- R8, R80-R84 - 10-kilo-ohm
- R85 - 10-ohm, 2W

Capacitors:

- C1, C2 - 33pF ceramic disk
- C3-C5 - 0.01µF ceramic disk
- C6 - 1000µF, 35V electrolytic
- C7, C9 - 0.1µF ceramic disk
- C8 - 470µF, 25V electrolytic

Miscellaneous:

- X1 - 230V AC primary to 9V, 1A secondary transformer
- X_{cr} - 32.768kHz crystal
- S1-S5, S10-S13 - Push-to-on (tactile) switch
- S6-S9 - SPDT switch
- S14 - Toggle switch
- IC base - IC base
- Heat-sink